DormDesk

Role-Based Hostel Complaint Management System

Project Documentation Report

Project Name	DormDesk
Subtitle	Role-Based Hostel Complaint Management System
Role	Cloud Backend Architect & Full-Stack Developer
Tech Stack	AWS (Cognito, Lambda, EC2), Node.js, Express, React, Tailwind CSS, Docker, Nginx, PostgreSQL/MongoDB

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1 Problem Statement & Motivation

Traditional hostel complaint handling is often fragmented. Students rely on physical registers, informal WhatsApp groups, or handwritten slips to report issues. Those channels create a predictable set of problems: complaints get lost, status updates are inconsistent, ownership is unclear, and there is little accountability for maintenance teams or hostel administrators.

DormDesk addresses that gap by providing a centralized, transparent, full-stack complaint lifecycle platform. Every complaint moves through a controlled workflow from submission to resolution, with access restricted by role and every administrative action recorded for traceability. The result is a system that reduces friction for students, improves visibility for hostel staff, and creates a dependable audit trail for management.

2 System Architecture & Core Features

DormDesk is designed as a role-aware platform with a clear separation between the user interface, API layer, and cloud infrastructure. It combines a React-based frontend, a Node.js and Express backend, and deployment components suited for containerized cloud hosting.

2.1 Role-Based Access Control (RBAC)

Access is divided across multiple user categories to keep the system secure and focused:

- **Students** can register, submit complaints, and track their own tickets.
- **Super Admins** can view and manage all complaints across categories.
- **Department-Specific Admins** can work only within their assigned domains, including Food, Water, Electrical, Cleaning, and Room.

This model prevents unauthorized access while keeping each role's interface simple and relevant.

2.2 Lifecycle Management

Each complaint moves through a four-stage visual lifecycle:

Received → Acknowledged → In Progress → Resolved

This stage model makes progress easy to understand at a glance and supports structured escalation and closure.

2.3 Accountability & Auditing

DormDesk records every meaningful administrative transition in an audit log. That includes the stage change, the person who performed it, the note attached to the action, and the timestamp. This creates visibility into what happened, when it happened, and who made the decision.

2.4 Cloud Infrastructure

The deployment model uses Docker containers behind an Nginx reverse proxy, with AWS services supporting authentication and automation. Cognito handles identity and group-based access, while Lambda is available for background triggers and future workflow extensions. This

architecture is production-oriented and can be adapted to either single-host deployment or cloud-hosted scaling patterns.

3 Visual Walkthrough & Implementation

The following screenshots document the primary user journey and the administrative control surfaces of the system.

3.1 Authentication Interface

Figure 1 shows the clean login screen and the local test credential panel. This view supports fast role switching during development and demonstrates the authentication entry point for students and administrators.

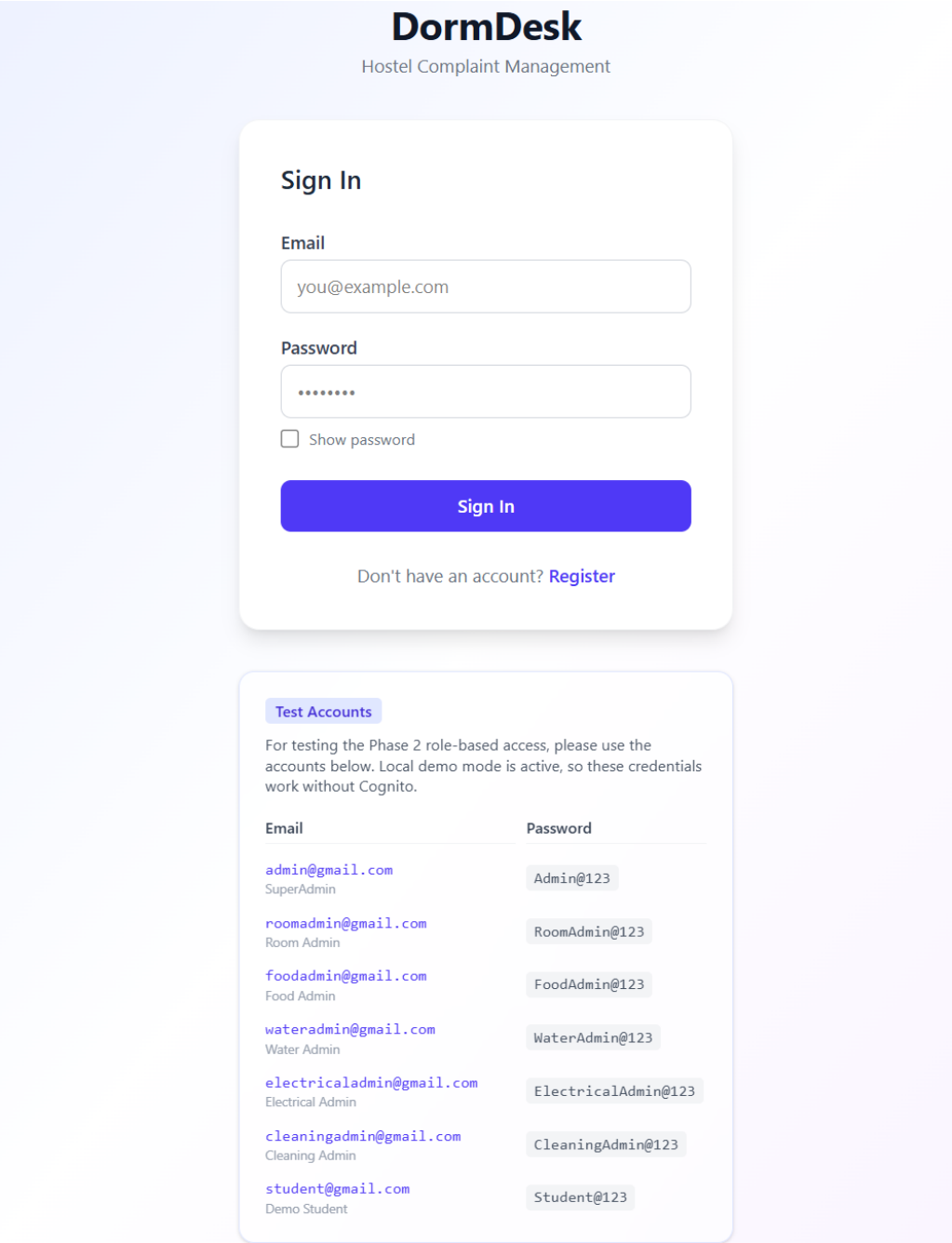


Figure 1: Authentication interface with local test credentials for role-based testing.

3.2 Student Complaint Dashboard

Figure 2 highlights the student workflow. The complaint form captures category, description, and priority, while the complaint list shows real-time visual progress for submitted tickets.

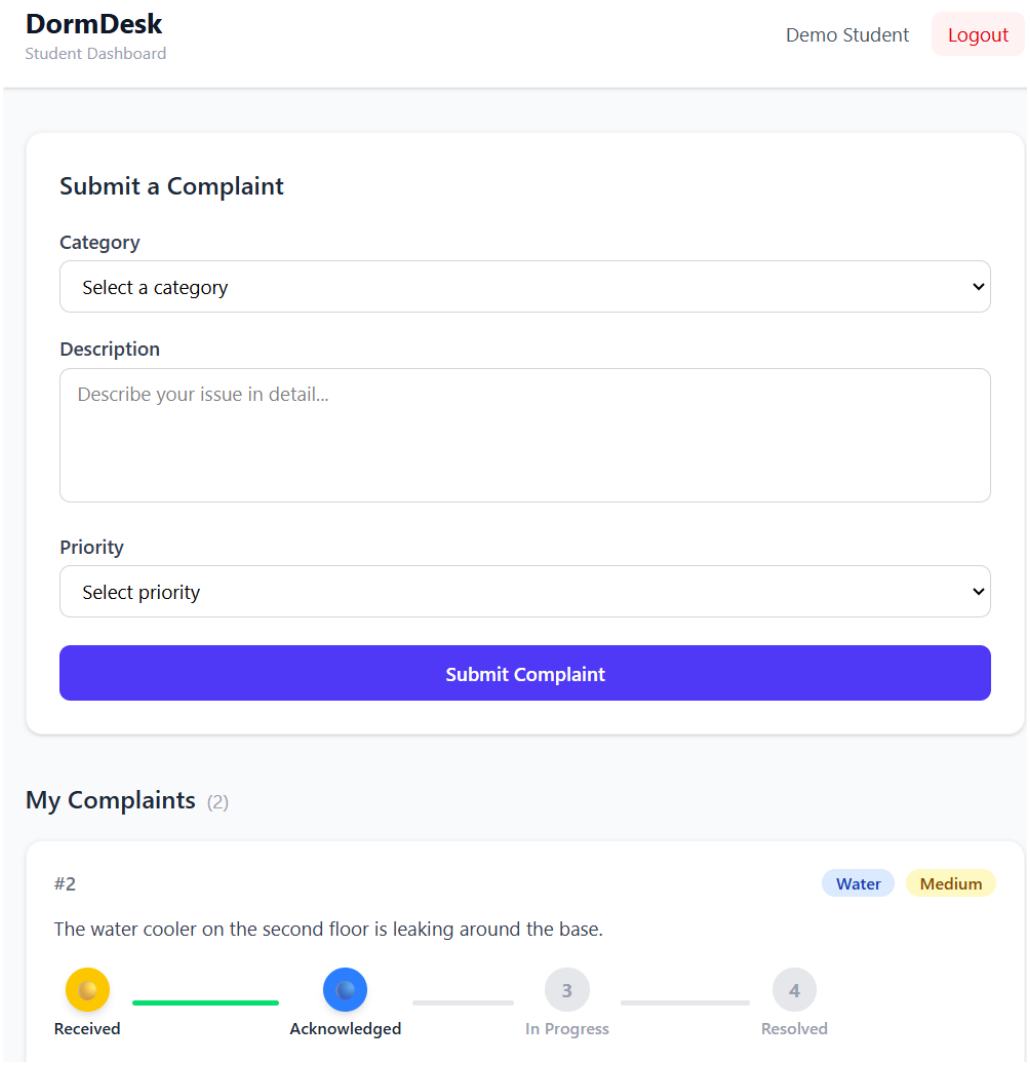
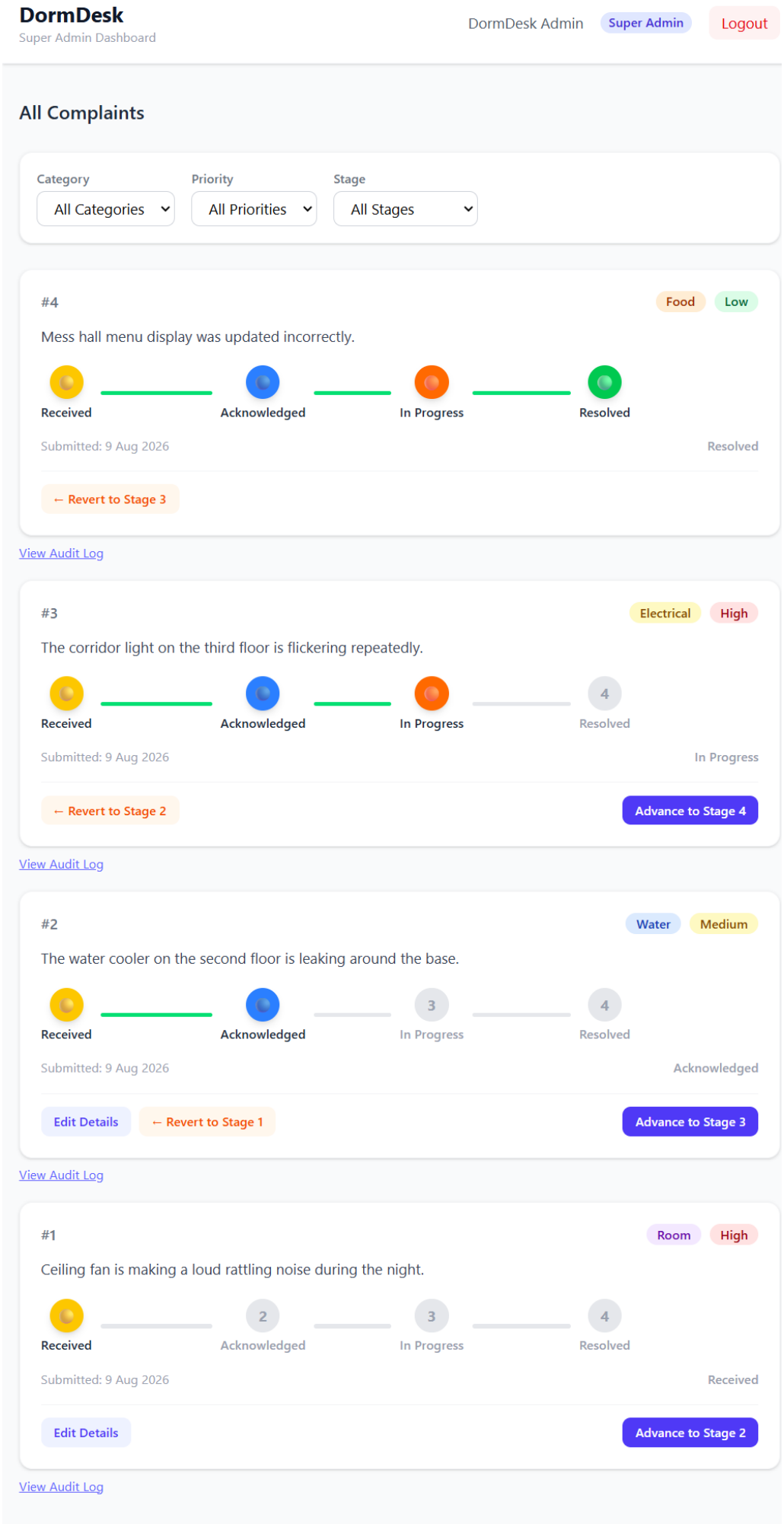


Figure 2: Student dashboard with complaint submission and live status tracking.

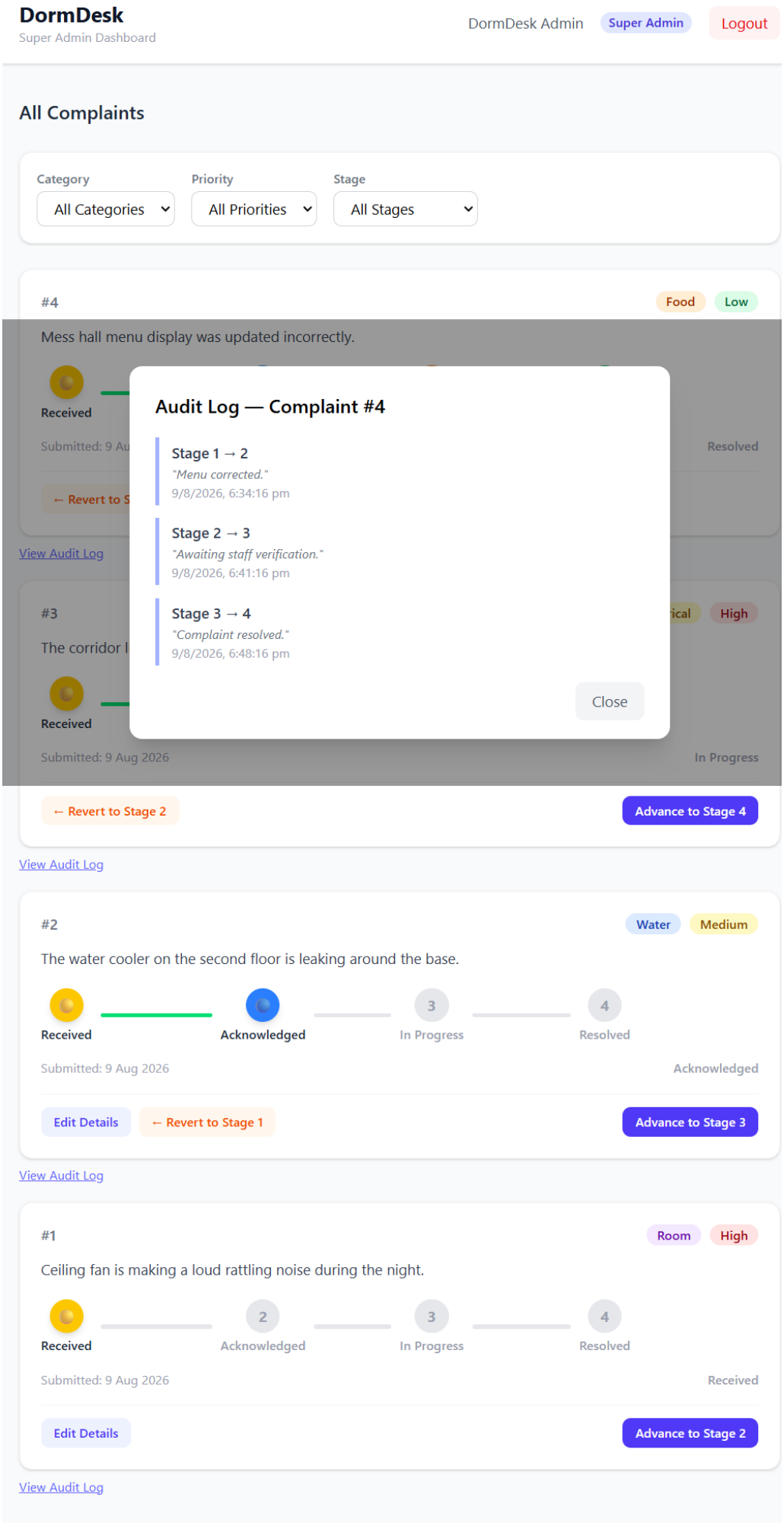
3.3 Admin Management Console

Figure 3 shows the admin control panel. It provides granular filters by category, priority, and stage, along with action buttons that allow administrators to advance or revert complaints through the workflow.


DormDesk
Super Admin Dashboard
DormDesk Admin
Super Admin
Logout
All Complaints
Category
All Categories
Priority
All Priorities
Stage
All Stages
#4
Food
Low
Mess hall menu display was updated incorrectly.
Received
Acknowledged
In Progress
Resolved
Submitted: 9 Aug 2026
Resolved
← Revert to Stage 3
View Audit Log
#3
Electrical
High
The corridor light on the third floor is flickering repeatedly.
Received
Acknowledged
In Progress
4
Resolved
Submitted: 9 Aug 2026
In Progress
← Revert to Stage 2
Advance to Stage 4
View Audit Log
#2
Water
Medium
The water cooler on the second floor is leaking around the base.
Received
Acknowledged
3
In Progress
4
Resolved
Submitted: 9 Aug 2026
Acknowledged
Edit Details
← Revert to Stage 1
Advance to Stage 3
View Audit Log
#1
Room
High
Ceiling fan is making a loud rattling noise during the night.
Received
2
Acknowledged
3
In Progress
4
Resolved
Submitted: 9 Aug 2026
Received
Edit Details
Advance to Stage 2
View Audit Log

3.4 Audit Log Transparency Layer

Figure 4 illustrates the accountability layer. The modal lists timestamped history entries for each administrative change, making the lifecycle of a complaint fully traceable.


DormDesk
Super Admin Dashboard
DormDesk Admin
Super Admin
Logout
All Complaints
Category
All Categories
Priority
All Priorities
Stage
All Stages
#4
Food
Low
Mess hall menu display was updated incorrectly.
Received
Submitted: 9 Au
Resolved
← Revert to S
View Audit Log
Audit Log — Complaint #4
Stage 1 → 2
"Menu corrected."
9/8/2026, 6:34:16 pm
Stage 2 → 3
"Awaiting staff verification."
9/8/2026, 6:41:16 pm
Stage 3 → 4
"Complaint resolved."
9/8/2026, 6:48:16 pm
Close
#3
ical
High
The corridor l
Received
Submitted: 9 Aug 2026
In Progress
← Revert to Stage 2
Advance to Stage 4
View Audit Log
#2
Water
Medium
The water cooler on the second floor is leaking around the base.
Received
Acknowledged
3
In Progress
4
Resolved
Submitted: 9 Aug 2026
Acknowledged
Edit Details
← Revert to Stage 1
Advance to Stage 3
View Audit Log
#1
Room
High
Ceiling fan is making a loud rattling noise during the night.
Received
2
Acknowledged
3
In Progress
4
Resolved
Submitted: 9 Aug 2026
Received
Edit Details
Advance to Stage 2
View Audit Log

4 Conclusion & Business Value

DormDesk eliminates the communication gaps that commonly affect institutional housing operations. Instead of relying on informal updates or paper-based processes, the platform centralizes complaints, assigns clear role boundaries, and preserves a complete record of every change.

From a technical perspective, the project demonstrates a production-ready cloud architecture with containerization, reverse proxying, role-based authentication, and an auditable workflow model. From a business perspective, it improves maintenance responsiveness, reduces confusion, and gives hostel management a reliable operational record that can be used for reporting, accountability, and service improvement.

Project Outcome: DormDesk provides a scalable foundation for digital hostel operations and can be extended with notifications, analytics, automated escalation, and deeper cloud integrations.